

Ordinary Differential Equation

§1.1 基本概念

1. 存在性定理: 设 $D = \{(x, y) \mid |x - x_0| \leq a, |y - y_0| \leq b\}$

$f(x, y)$ 在 D 上连续, 且一阶偏导数 $f'_y(x, y)$ 连续

则方程 $\begin{cases} y' = f(x, y) \\ y(x_0) = y_0 \end{cases}$ 在 $|x - x_0| \leq \alpha$ 上有唯一解.

其中 $\alpha = \min\{a, \frac{b}{M}\}$, $M = \max|f(x, y)|$

初值条件是一个定解条件 (确定通解为特解)

2. 主要步骤: 建模、求解、分析 (确定特解, 修正模型等)

§1.2 可分离变量方程: 齐次方程

1. 可分离变量方程: $\begin{cases} \frac{dy}{dx} = f(x, y) = \varphi(x)\psi(y) \\ y|_{x=x_0} = y_0 \end{cases}$

① 若 $\psi(y^*) = 0$, 则 $y(x) = y^*$ 是解.

② 若 $\psi(y) \neq 0$, 则 $\frac{dy}{\psi(y)} = \varphi(x)dx$, $\int \frac{dy}{\psi(y)} = \int \varphi(x)dx + C$ 是方程的通解.

若 $y|_{x=x_0} = y_0$, 代入方程 $\int_{y_0}^y \frac{dt}{\psi(t)} = \int_{x_0}^x \varphi(t)dt + C \Rightarrow C = ?$. 求出特解.

例: Logistic 人口模型 $\begin{cases} \frac{dP}{dt} = kP(1 - \frac{P}{M}) \\ P|_{t=0} = P_0 \end{cases}$

解: ① $\psi(M) = 0$. 故 $y = M$ 是解.

② $\psi(P) \neq 0$, 则 $\frac{dP}{P(1 - \frac{P}{M})} = kdt \Rightarrow \int \frac{M dP}{P(M - P)} = \int kdt + C$
 $\Rightarrow \int (\frac{1}{P} + \frac{1}{M - P}) dP = kt + C \Rightarrow \ln|\frac{P}{M - P}| = kt + C \Rightarrow \frac{P}{M - P} = \pm e^C e^{kt}$, 令 $A = \pm e^C$.
即 $\frac{P}{M - P} = A e^{kt}$, $P = \frac{M}{A e^{-kt} + 1}$. 代入 $P(0) = P_0$, 得 $P = \frac{M P_0}{(M - P_0) e^{-kt} + P_0}$.

2. 齐次方程: $\begin{cases} \frac{dy}{dx} = g(\frac{y}{x}) \\ y|_{x=x_0} = y_0 \end{cases}$ 称为零次齐次方程
若 $f(\lambda x, \lambda y) = \lambda^k f(x, y)$, 则称 f 为 k 次齐次函数.

令 $u = \frac{y}{x}$, 则 $y = ux \Rightarrow y' = u'x + u = g(u) \Rightarrow \frac{dy}{dx} = \frac{1}{x}(g(u) - u)$

例: $\frac{dy}{dx} = \frac{x+y}{x-y}$

解: $\frac{dy}{dx} = 1 + (\frac{y}{x}) / [1 - (\frac{y}{x})]$

令 $u = \frac{y}{x}$, 则 $y = ux$, $y' = u'x + u = \frac{1+u}{1-u} \Rightarrow u' = \frac{1}{x} \cdot \frac{1+u^2}{1-u}$

$\Rightarrow \frac{(1+u)du}{1+u^2} = \frac{1}{x} dx \Rightarrow \int \frac{(1+u)du}{1+u^2} = \int \frac{1}{x} dx + C$

$\Rightarrow \int \frac{1}{1+u^2} du - \frac{1}{2} \int \frac{1}{1+u^2} du = \ln|x| + C$

$\Rightarrow \arctan u - \frac{1}{2} \ln|1+u^2| = \ln|x| + C \Rightarrow \arctan u = \ln|x\sqrt{1+u^2}| + C$.

代入 $u = \frac{y}{x}$, 得 $\arctan \frac{y}{x} = \ln\sqrt{x^2 + y^2} + C$.

取对数, 得 $\sqrt{x^2 + y^2} = A e^{\arctan \frac{y}{x}}$.

例: $\frac{dy}{dx} = \frac{x+y+1}{x-y+2}$.

解: 设 $\xi + \eta = x + y + 1$, $\xi - \eta = x - y + 2$, 则 $\sqrt{\xi^2 + \eta^2} = A e^{\arctan \frac{\eta}{\xi}}$.

$\xi = x + \frac{3}{2}$, $\eta = y - \frac{1}{2}$, $\frac{d\eta}{d\xi} = \frac{dy}{dx}$.

以下解法同上例.

§1.3 一阶线性微分方程 · 伯努利方程

1. 一阶线性微分方程

一般形式: $\frac{dy}{dx} + p(x)y = f(x)$

当 $f(x) \equiv 0$ 时, 称为一阶线性齐次方程: $\frac{dy}{dx} = -p(x)y$

① $y=0$ 是解.

② $y \neq 0$ 时, $\frac{dy}{y} = -p(x)dx \Rightarrow \int \frac{dy}{y} = -\int p(x)dx + C$

$\Rightarrow \ln|y| = -\int p(x)dx + C \Rightarrow |y| = e^{-\int p(x)dx} \cdot e^C$

故方程的通解为 $y = ce^{-\int p(x)dx}$, $c \in \mathbb{R}$.

当 $f(x) \neq 0$ 时, 采用变动任意常数法.

从对应的齐次方程出发, 变动常数设 $y(x) = u(x)e^{-\int p(x)dx}$

两边求导, 得 $\frac{dy}{dx} = \frac{du}{dx}e^{-\int p(x)dx} - up(x)e^{-\int p(x)dx}$

代入, 得 $e^{-\int p(x)dx} \frac{du}{dx} - ue^{-\int p(x)dx} p(x) + p(x)ue^{-\int p(x)dx} = f(x)$

即 $e^{-\int p(x)dx} \frac{du}{dx} = f(x)$, $u = \int f(x)e^{\int p(x)dx} dx + C$.

将不定积分改为定积分, 得 $y = e^{-\int_{x_0}^x p(t)dt} \left[\int_{x_0}^x f(t)e^{\int_{x_0}^t p(t)dt} du + y_0 \right]$.

$x = x_0$ 时, $y = y_0$.

设 $Y(x)$ 是齐次的通解, y^* 是非齐次的特解, 则 $y = Y + y^*$.

2. 伯努利方程

一般形式: $\frac{dy}{dx} + p(x)y = f(x)y^n$, n 是常数.

① $y=0$ 是解.

② 两边同时除以 y^n , 得 $y^{-n} \frac{dy}{dx} + p(x)y^{1-n} = f(x)$

令 $z = y^{1-n}$, 得 $\frac{dz}{dx} + (1-n)p(x)z = (1-n)f(x)$, 解法同 一阶线性方程.

例. (书 P40 例1)

解: 设 $y = y(x)$, 设交点为 $(0, Y^*)$, 切点为 (x, y) .

由 $|OA| = |AM|$ 得 $(Y^*)^2 = x^2 + y^2 - 2yY^* + (Y^*)^2 \Rightarrow x^2 + y^2 - 2yY^* = 0$.

AM: $y - Y^* = y'x \Rightarrow Y^* = y - y'x$

代入得 $x^2 + y^2 - 2y(y - y'x) = 0$, 得 $x^2 - y^2 + 2yy'x = 0$.

写成标准形求得 $\frac{dy^2}{dx} - \frac{y^2}{x} = -x$.

$$\text{令 } z = y^2, \quad \frac{dz}{dx} - \frac{1}{x}z = -x$$

$$\text{通解 } z = e^{\int \frac{1}{x} dx} [C + \int (-x) e^{-\int \frac{1}{x} dx} dx] = e^{\ln|x|} [C + \int (-x) \frac{1}{|x|} dx]$$

$$= x(C-x) \quad * \text{也可用变量常数法.}$$

$$\text{故 } (x - \frac{3}{2})^2 + y^2 = (\frac{3}{2})^2.$$

$$\text{代入点 } (\frac{3}{2}, \frac{3}{2}) \text{ 得 } C=3. \text{ 故所求的方程为 } (x - \frac{3}{2})^2 + y^2 = (\frac{3}{2})^2, \quad 0 < x < 3.$$

§1.4 全微分方程

1. 全微分方程

标准形式: $M(x, y)dx + N(x, y)dy = 0$.

$$du(x, y) = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

若 $du(x, y)$, 则 $u(x, y) = C$.

· 如果是全微分方程, 则 $\exists u(x, y)$, 有

$$\frac{\partial u}{\partial x} = M(x, y), \quad \frac{\partial u}{\partial y} = N(x, y)$$

$$\Rightarrow \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right)$$

定理: 设 $G \in \mathbb{R}^2$ 是单连通的, 且有连续的一阶偏导数, 则方程是全微分方程的必要条件为

$$\frac{\partial M}{\partial y} \equiv \frac{\partial N}{\partial x} \quad (\text{二阶混合偏导数相同, 证明请见数学分析笔记})$$

$$\text{解方程: } \frac{\partial u}{\partial x} = M(x, y) \Rightarrow u(x, y) = \int M(x, y) dx + \varphi(y)$$

$$\frac{\partial u}{\partial y} = N(x, y) \Rightarrow \varphi'(y) = N(x, y) - \frac{\partial}{\partial y} \int M(x, y) dx$$

构造全微分的技巧:

① 分离出 $M(x)dx + N(y)dy$ 的部分

② 凑微分

$$\text{例. } \frac{dy}{dx} = -\frac{1+xy^2}{1+x^2y}, \quad y(0)=1.$$

$$\text{解: } (1+xy^2)dx + (1+x^2y)dy = 0.$$

$$\text{则 } \frac{\partial M}{\partial y} = 2xy, \quad \frac{\partial N}{\partial x} = 2xy. \text{ 故上式为全微分方程.}$$

$$\text{方法一: } \frac{\partial u}{\partial x} = M = 1+xy^2 \Rightarrow u(x, y) = \int (1+xy^2) dx + \varphi(y)$$

$$u(x, y) = x + \frac{1}{2}x^2y^2 + \varphi(y)$$

$$\frac{\partial u}{\partial y} = x^2y + \varphi'(y) = 1 + x^2y \Rightarrow \varphi'(y) = 1, \text{ 可令 } \varphi(y) = y.$$

$$\text{故 } u(x, y) = x + y + \frac{1}{2}x^2y^2, \text{ 原方程通解: } x + y + \frac{1}{2}x^2y^2 = c.$$

$$\text{方法二: } dx + dy + (xy^2 dx + x^2y dy) = 0.$$

$$\Rightarrow dx + dy + d\frac{x^2y^2}{2} = 0.$$

$$\Rightarrow x + y + \frac{1}{2}x^2y^2 = c.$$

· 若 $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, 但存在非零函数 $\mu(x, y)$, 使得

$$\frac{\partial(\mu M)}{\partial y} = \frac{\partial(\mu N)}{\partial x}, \text{ 则称 } \mu \text{ 是原方程的一个积分因子.}$$

$$\frac{\partial \mu}{\partial y} M + \mu \frac{\partial M}{\partial y} = \frac{\partial \mu}{\partial x} N + \mu \frac{\partial N}{\partial x}$$

$$\text{若 } \mu \equiv \mu(x), \text{ 得 } \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} \equiv \varphi(x) = \frac{1}{\mu} \cdot \frac{d\mu}{dx}$$

$$\text{若 } \mu \equiv \mu(y), \text{ 得 } \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} \equiv \psi(y) = \frac{1}{\mu} \cdot \frac{d\mu}{dy}$$

$$\text{例. } y dx + (y^2 - 2x) dy = 0$$

$$\text{解: 显然 } \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}.$$

$$\text{设积分因子为 } \mu, \text{ 则 } \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{-M} = \frac{3}{-y} = \frac{1}{\mu} \cdot \frac{d\mu}{dy}.$$

$$\text{取 } \mu(y) = \frac{1}{y^3}, \text{ 则 } \frac{1}{y^3} dx + \frac{1}{y} dy - \frac{x}{y^3} dy = 0$$

$$\text{即 } d\left(\frac{x}{y^3}\right) + d(\ln|y|) = 0, \frac{x}{y^3} + \ln|y| = c.$$

另解: 可化为一阶非齐次线性方程.

常见的一些可凑全微分的二元函数:

$$y dx + x dy = d(xy), \frac{y dx - x dy}{y^2} = d\left(\frac{x}{y}\right), \frac{x dy - y dx}{x^2} = d\left(\frac{y}{x}\right)$$

$$\frac{x dy - y dx}{x^2 + y^2} = d(\arctan \frac{y}{x}), \frac{y dx - x dy}{x^2 - y^2} = d\left(\frac{1}{2} \ln \left| \frac{x-y}{x+y} \right| \right).$$

§1.5 可降阶的二阶微分方程

利用变量变换转化为一阶方程

$$\text{一般形式: } \frac{d^2 y}{dx^2} = f(x, y, \frac{dy}{dx}), y|_{x=x_0} = y_0, \frac{dy}{dx}|_{x=x_0} = y'_0$$

通解中未知数的阶数 = 方程的阶数

$$1. \frac{d^2 y}{dx^2} = f(x) \Rightarrow \frac{dy}{dx} = \int f(x) dx + c_1 \Rightarrow y = \int [\int f(x) dx + c_1] dx + c_2.$$

$y^{(n)}(x) = f(x)$ 则积分 n 次

$$2. \frac{d^2 y}{dx^2} = f(x, \frac{dy}{dx})$$

$$\text{令 } \frac{dy}{dx} = p, \text{ 则 } \frac{d^2 y}{dx^2} = \frac{dp}{dx} \Rightarrow \frac{dp}{dx} = f(x, p)$$

$$\text{则 } p = \varphi(x, c_1), \frac{dy}{dx} = \varphi(x, c_1) \Rightarrow y = \int \varphi(x) dx + c_2.$$

$$3. \frac{d^2 y}{dx^2} = f(y, \frac{dy}{dx})$$

$$\text{令 } \frac{dy}{dx} = p, \text{ 则 } \frac{d^2 y}{dx^2} = \frac{dp}{dx} \Rightarrow \frac{dp}{dx} = f(y, p)$$

$$dx = \frac{1}{p} dy, \text{ 则 } \frac{dp}{dx} = p \frac{dp}{dy} \quad \text{或} \quad \frac{dp}{dx} = \frac{dp}{dy} \cdot \frac{dy}{dx} = p \frac{dp}{dy}.$$

$$\text{原方程化为 } p \frac{dp}{dy} = f(y, p) \Rightarrow p = \varphi(y, c_1)$$

$$\frac{dy}{dx} = \varphi(y, c_1) \quad \text{初值: } p(y_0) = p_0.$$

$$4. y^{(n)}(x) = f(x, y^{(n-1)}(x))$$

$$\text{令 } y^{(n-1)}(x) = p, \text{ 则 } y^{(n)}(x) = \frac{dp}{dx}.$$

$$\text{例: } y'' = (1-2y)(y')^3 - (y')^2, \text{ 且 } y(1) = 0, y'(1) = -1.$$

$$\text{解: 令 } y' = p, \text{ 则 } y'' = p \frac{dp}{dy}$$

$$\text{故 } p \frac{dp}{dy} = (1-2y)p^3 - p^2$$

$$\textcircled{1} p = 0 \text{ 时, } y'(x) = 0, \text{ 故舍去.}$$

$$\textcircled{2} \frac{dp}{dy} = (1-2y)p^2 - p, \text{ 即 } \frac{dp}{dy} + p = (1-2y)p^2. \text{ (伯努利方程)}$$

$$\text{令 } z = \frac{1}{p}, \frac{dz}{dy} = -\frac{1}{p^2} \cdot \frac{dp}{dy} = -\frac{1}{p^2} (1-2y)p^2 + \frac{1}{p^2} \cdot p = (2y-1) + z.$$

$$\Rightarrow \frac{dz}{dy} - z = 2y-1.$$

$$\text{解得 } z(y) = e^y (c_1 + \int (2y-1)e^{-y} dy) = e^y (c_1 - e^{-y}(2y-1) - 2e^{-y}) = ce^y - 2y - 1.$$

$$\text{故 } p = \frac{1}{ce^y - 2y - 1}, \text{ 当 } y = 0 \text{ 时, } p = \frac{dy}{dx} = -1.$$

$$\text{代入, 解得 } c = 0, p = -\frac{1}{2y+1}.$$

$$\frac{dy}{dx} = -\frac{1}{2y+1} \Rightarrow (2y+1)dy = -dx \Rightarrow y^2 + y = -x + c'.$$

$$\text{当 } x=1 \text{ 时, } y=0, \text{ 代入得 } c' = 1$$

$$\text{综上, 得 } y^2 + y = 1 - x.$$

§2.1 线性微分方程解的一般理论

一般形式: $\frac{d^ny}{dx^n} + p_1(x) \frac{d^{n-1}y}{dx^{n-1}} + \dots + p_{n-1}(x) \frac{dy}{dx} + p_n(x)y \equiv L_n[y] = f(x)$

若 $f(x) \equiv 0 \Rightarrow$ 齐次线性方程

需要有 n 个初值条件, 即 $y^{(m)}(x_0) = y_0^{(m)}$.

定理: 若 $p_i(x)$ 与 $f(x)$ 是连续的, 且具有 n 个初值, 则存在唯一的 $y = \varphi(x)$ 满足条件.

1. $L[y] \equiv 0$.

引理 1: $L[cy] = cL[y]$

引理 2: $L[y_1 + y_2] = L[y_1] + L[y_2]$

引理 3: $L[\sum_{i=1}^n C_i y_i] = \sum_{i=1}^n C_i L[y_i]$

$L[y]$ 称为线性微分算子.

定义: 线性相关(无)关.

设 $y_i(x)$ ($i=1, 2, \dots, n$) 定义在 D , 若存在 n 个不全为 0 的常数, 使得 $\forall x \in D$, 有

$$C_1 y_1(x) + C_2 y_2(x) + \dots + C_n y_n(x) \equiv 0,$$

则称 $y_i(x)$ 在 D 上线性相关, 否则线性无关.

定义: 朗斯基行列式

$$W(x) = \begin{vmatrix} y_1(x) & y_2(x) & \dots & y_n(x) \\ y_1'(x) & y_2'(x) & \dots & y_n'(x) \\ \vdots & \vdots & & \vdots \\ y_1^{(n-1)}(x) & y_2^{(n-1)}(x) & \dots & y_n^{(n-1)}(x) \end{vmatrix}$$

定理: 齐次线性方程不同解的线性组合仍是它的解.

即若 $y_i(x)$ 是 $L[y] = 0$ 的解, 则 $L[y_1, y_2, \dots, y_n] = 0$.

定理: 不同解之间线性相关的充要条件是它们的朗斯基行列式 $W(x) \equiv 0$.

线性无关的充要条件是 $W(x) \neq 0, \forall x$.

以二阶为例: $y''(x) + p_1(x)y' + p_2(x)y = 0$, 有两个解 $y_1(x)$ 与 $y_2(x)$.

$$\text{则 } W(x) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix}, \text{ 则 } W'(x) = \begin{vmatrix} y_1'(x) & y_2'(x) \\ y_1''(x) & y_2''(x) \end{vmatrix} + \begin{vmatrix} y_1(x) & y_2(x) \\ y_1''(x) & y_2''(x) \end{vmatrix}$$

$$\text{而 } y'' = -p_1(x)y' - p_2(x)y,$$

$$\text{故 } W = \begin{vmatrix} y_1 & y_2 \\ -p_1 y_1' - p_2 y_1 & -p_1 y_2' - p_2 y_2 \end{vmatrix} = -p_1 \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = -p_1 W(x)$$

$$\text{即 } \frac{dw}{dx} = -p_1 w, \quad w(x) = w(x_0) e^{-\int p_1(x) dx}$$

即当 $x_0 = 0$ 时, $w(x) \equiv 0$.

推论: 若 $\exists x_0$, 使得 $w(x_0) = 0$, 则 $y_i(x)$ 线性相关; 反之, 则线性无关.

定理: n 阶线性方程若有 n 个线性无关的解, 则它们的线性组合即为通解.

定理: n 阶齐次方程必有 n 个线性无关的解.

$$2. L[y] = f(x)$$

设 y^* 是 $L[y] = f(x)$ 的特解, $y_i(x)$ 是 $L[y] \equiv 0$ 的通解.

则 $L[y] = f(x)$ 的通解为 $y = L(y_1, y_2, \dots, y_n) + y^*$.

§2.2 常系数线性微分方程的解法

$$\text{一般形式: } y^{(n)}(x) + p_1 y^{(n-1)}(x) + \dots + p_n y(x) = 0.$$

$$1. \text{一阶 } y' = ay \Rightarrow y = e^{ax}$$

$$\text{设 } y = e^{\lambda x}, \text{ 则得到一个关于 } \lambda \text{ 的多项式 } D(\lambda) = \lambda^n + p_1 \lambda^{n-1} + \dots + p_{n-1} \lambda + p_n = 0.$$

称为 $L[y]$ 的特征方程.

代数基本定理: $D(\lambda)$ 必有 n 个复数解.

设解为 $\lambda_i (i=1, \dots, n)$, 则有 n 个解 $y = e^{\lambda_i x}$.

$$2. \text{二阶 } y'' + p y' + q y = 0$$

$$\text{特征方程 } \lambda^2 + p\lambda + q = 0, \text{ 特征值 } \lambda = \frac{-p \pm \sqrt{p^2 - 4q}}{2}$$

① $\sqrt{p^2 - 4q} \neq 0$, 则有两个根 λ_1, λ_2 .

$W(x) = e^x (\lambda_1 - \lambda_2) \neq 0$, 故 y_1 与 y_2 线性无关.

故通解 $y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$.

当 λ_1, λ_2 是复数时, $\lambda = \alpha \pm i\beta$

$$\text{故通解 } y = c_1 e^{(\alpha+i\beta)x} + c_2 e^{(\alpha-i\beta)x} = e^{\alpha x} (c_1 e^{i\beta x} + c_2 e^{-i\beta x})$$

令实部 $R(x) = e^{\alpha x} \cos \beta x$, 虚部 $I(x) = e^{\alpha x} \sin \beta x$ ($\beta \neq 0$).

故通解为 $y = c_1 e^{\alpha x} \cos \beta x + c_2 e^{\alpha x} \sin \beta x$.

② $\lambda_1 = \lambda_2 = \lambda$ 时, $y = e^{\lambda x}$ 是一个解.

则 $y'' + py' + \frac{p^2}{4}y = 0$, 设 $y = ue^{\lambda x}$

则 $u''e^{\lambda x} + 2\lambda u'e^{\lambda x} + \lambda^2 u e^{\lambda x} + pu'e^{\lambda x} + \lambda p u e^{\lambda x} + \frac{p^2}{4}e^{\lambda x} = 0$.

由于 $\lambda = -\frac{p}{2}$, 且 $\lambda^2 e^{\lambda x} + p\lambda e^{\lambda x} + \frac{p^2}{4}e^{\lambda x} = 0$.

故 $u''e^{\lambda x} = 0$, 即 $u = c_1 x + c_2$.

故 $[Ly]$ 的通解为 $y = c_1 x e^{\lambda x} + c_2 e^{\lambda x}$.

3. n 阶 $(\lambda - \lambda_1)^{n_1} (\lambda - \lambda_2)^{n_2} \dots (\lambda - \lambda_s)^{n_s} = 0$

单根对应单解 $e^{\lambda x}$, 重根对应重解 $e^{\lambda x}, x e^{\lambda x}, \dots, x^{m-1} e^{\lambda x}$. (m 为 λ 的重数)

例: $\ddot{s} - 4\dot{s} + 4s = 0, s(0) = 0, \dot{s}(0) = 2$.

解: 特征方程 $\lambda^2 - 4\lambda + 4 = 0$, 则有重根 $\lambda = 2$.

故 $s = c_1 e^{2x} + c_2 x e^{2x}$, 代入初值条件得 $c_1 = 0, c_2 = 2$

故 $s = 2x e^{2x}$

例: $y'' + 4y' + 13y = 0$.

解: 特征方程 $\lambda^2 + 4\lambda + 13 = 0$, 有复根 $\lambda_1 = -2 + 3i$ 与 $\lambda_2 = -2 - 3i$

故 $y = e^{-2x} (c_1 \cos 3x + c_2 \sin 3x) = A e^{-2x} \sin(3x + \varphi)$.

4. 非齐次解法 $L[y] = f(x)$.

① $f(x) = p_m(x) e^{\alpha x}$ 时, (p_m 表示 m 阶多项式)

猜测 $y^* = Q(x) e^{\alpha x}$, $Q(x)$ 也是多项式

二阶时 $Q''(x) e^{\alpha x} + 2\alpha Q'(x) e^{\alpha x} + \alpha^2 Q(x) e^{\alpha x} + pQ'(x) e^{\alpha x} + q\alpha Q(x) e^{\alpha x} + qQ(x) e^{\alpha x} = p_m e^{\alpha x}$

则 $Q''(x) + (2\alpha + p)Q'(x) + (\alpha^2 + p\alpha + q)Q(x) = p_m(x)$

若 α 不是特征值时, 则 $\alpha^2 + p\alpha + q \neq 0$, 故 $Q(x)$ 也是 m 阶多项式.

若 α 是单根, 则 $\alpha^2 + p\alpha + q = 0, 2\alpha + p \neq 0$, 故 $Q(x)$ 是 $(m+1)$ 阶多项式, 且 $y^* = x Q_m(x)$

若 α 是二重根, 则 $\alpha^2 + p\alpha + q = 2\alpha + p = 0$, 故 $Q(x)$ 是 $(m+2)$ 阶多项式, 且 $y^* = x^2 Q_m(x)$

例: $y^{(4)} - 2y''' + 3y'' - 2y' + y = 0$

解: 特征方程 $\lambda^4 - 2\lambda^3 + 3\lambda^2 - 2\lambda + 1 = 0$.

$$\Rightarrow (\lambda^2 + 1)^2 + \lambda^2 - 2\lambda(\lambda^2 + 1) = 0 \Rightarrow (\lambda^2 + 1 - \lambda)^2 = 0.$$

故原方程有二重根 $\lambda = \frac{1}{2} \pm \frac{\sqrt{3}}{2}i$

故原方程的通解为 $y_1 = e^{\frac{1}{2} + \frac{\sqrt{3}}{2}i}x = e^{\frac{1}{2}x} (\cos \frac{\sqrt{3}}{2}x + i \sin \frac{\sqrt{3}}{2}x)$, $y_2 = e^{\frac{1}{2}x} (\cos \frac{\sqrt{3}}{2}x - i \sin \frac{\sqrt{3}}{2}x)$

实解为 $y_1 = e^{\frac{1}{2}x} \cos \frac{\sqrt{3}}{2}x$, $y_2 = e^{\frac{1}{2}x} \sin \frac{\sqrt{3}}{2}x$, $y_3 = x e^{\frac{1}{2}x} \cos \frac{\sqrt{3}}{2}x$, $y_4 = x e^{\frac{1}{2}x} \sin \frac{\sqrt{3}}{2}x$.

故原方程的解为 $e^{\frac{1}{2}x} [(c_1 + c_2x) \cos \frac{\sqrt{3}}{2}x + (c_3 + c_4x) \sin \frac{\sqrt{3}}{2}x]$

例: $y'' + 2y' + y = (x-5)e^{-x}$.

解: 特征方程 $\lambda^2 + 2\lambda + 1 = 0$

故齐次方程的通解为 $Y = (c_1 + c_2x)e^{-x}$.

设特解 $y^* = x^2(Ax+B)e^{-x}$, 代入方程, $(6Ax+2B)e^{-x} = (x-5)e^{-x}$.

故 $A = \frac{1}{6}$, $B = -\frac{5}{2}$, $y^* = x^2(\frac{1}{6}x - \frac{5}{2})e^{-x}$.

故原方程的通解为 $y = (\frac{1}{6}x^3 - \frac{5}{2}x^2 + c_1x + c_2)e^{-x}$.

5. 形如 $L[y] = P_m(x)e^{\alpha x} + Q_n(x)e^{\beta x}$ 的解

① 求 $L[y] = 0$ 的通解 Y .

② 求 $L[y] = P_m(x)e^{\alpha x}$ 的特解 y_1^*

③ 求 $L[y] = Q_n(x)e^{\beta x}$ 的特解 y_2^*

④ 则通解 $y = Y + y_1^* + y_2^*$.

6. 形如 $L[y] = P_m(x)e^{\alpha x} \cos bx$ 或 $Q_m(x)e^{\alpha x} \sin bx$ 的解.

① 求 $L[y] = 0$ 的通解 Y .

② 求 $L[y^*] = P_m(x)e^{(a+bi)x}$

③ $L[y] = P_m(x)e^{\alpha x} \cos bx$ 的解为 $Y + \operatorname{Re} y^*$.

另解.

① 求 $L[y] = 0$ 的通解 Y .

② 设 $a+bi$ 是 k 重根, 则设 $y^* = x^k e^{\alpha x} (R_m(x) \cos bx + H_m(x) \sin bx)$

③ 则 $y = Y + y^*$.

上述解法均可.

例: $y'' + y = \sin x - \cos 2x$.

解: 特征方程 $\lambda^2 + 1 = 0$, 则 $\lambda = \pm i$.

故齐次方程的通解为 $Y = C_1 \cos x + C_2 \sin x$

求解 $y'' + y = \sin x$, i 是单根

故设特解 $y_1^* = x(A \sin x + B \cos x)$, 则 $y_1^{*'} = (A - Bx) \sin x + (B + Ax) \cos x$,

$$y_1^{*''} = (-Ax - 2B) \sin x + (2A - Bx) \cos x$$

故 $y_1^{*''} + y_1^* = -2B \sin x + 2A \cos x = \sin x$, 故 $A = 0, B = -\frac{1}{2}$.

$$\text{故 } y_1^* = -\frac{1}{2} x \cos x$$

求解 $y'' + y = -\cos 2x$, $2i$ 不是根.

故设特解 $y_2^* = C \cos 2x + D \sin 2x$, 则 $y_2^* = \frac{1}{3} \cos 2x$.

故特解 $y^* = y_1^* + y_2^* = -\frac{1}{2} x \cos x + \frac{1}{3} \cos 2x$.

故原方程的解 $y = C_1 \cos x + C_2 \sin x - \frac{1}{2} x \cos x + \frac{1}{3} \cos 2x$.

§ 2.4 一般线性微分方程的一些解法

1. 变量变换法

欧拉方程: $a_0 x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_n y = f(x)$.

令 $x = e^t$, 则 $\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$, $t = \ln x$.

$$\text{故 } \frac{dy}{dx} = \frac{1}{x} \frac{dy}{dt}.$$

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{1}{x} \frac{dy}{dt} \right) = -\frac{1}{x^2} \frac{dy}{dt} + \frac{1}{x} \cdot \frac{d^2 y}{dt^2} \cdot \frac{dt}{dx} = \frac{1}{x^2} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right).$$

$$\text{则 } x^k \frac{d^k y}{dx^k} = D(D-1)\dots(D-k+1)y, D = \frac{d}{dt}. \quad \text{= 1阶时: } a_0 \frac{d^2 y}{dt^2} + (a_1 - a_0) \frac{dy}{dt} + a_2 y = f(e^t).$$

例: $x^2 y'' - 3xy' + 4y = x + x^2 \ln x$.

解: 令 $x = e^t$, 则 $\frac{d^2 y}{dt^2} - 4 \frac{dy}{dt} + 4y = e^t + te^{2t}$

① 特征方程 $\lambda^2 - 4\lambda + 4 = 0$, $\lambda = 2$.

故齐次方程的通解为 $(C_1 + C_2 t) e^{2t}$.

② 设 $y^* = Ae^t + t^2(Bt + C)e^{2t}$

解得 $y^* = e^t + \frac{1}{6} t^2 e^{2t}$

故原方程的解为 $y = (C_1 + C_2 t) e^{2t} + e^t + \frac{1}{6} t^3 e^{2t}$

将 $t = \ln x$ 代入, 得 $y = (C_1 + C_2 \ln x + \frac{1}{6} \ln^3 x) x^2 + x$.

可降阶的方程: $\frac{d^2 y}{dx^2} + p(x) \frac{dy}{dx} + q(x) = 0$.

通解为 $y = y_1(x) + y_2(x)$.

若已知非零解 y_1 , 作变量变换 $y = y_1 u$.

故 $y' = y_1' u + y_1 u'$, $y'' = y_1'' u + 2y_1' u' + y_1 u''$

代入方程, 得 $y_1 u'' + [2y_1' + p(x)y_1] u' = 0$ (由 $y'' + p(x)y' + q(x) = 0$ 可化简)

令 $z = u'$, 则 $\frac{z'}{z} = -\frac{2y_1'}{y_1} - p(x)$, 则 $\ln|z| = -2\ln|y_1| - \int p(x) dx + C$.

故 $z = \frac{C_2}{y_1^2} e^{-\int p(x) dx}$

故 $y = y_1 (C_1 + C_2 \int \frac{1}{y_1^2} e^{-\int p(x) dx} dx)$

例: $(1-x)y'' + xy' - y = 0$.

解: 猜测 $y = x$, 则易得是方程的特解.

化方程为 $y'' + \frac{x}{1-x} y' - \frac{1}{1-x} y = 0$.

直接用公式 $y = x (C_1 + C_2 \int \frac{1}{y^2} e^{-\int \frac{x}{1-x} dx} dx)$.

或设 $y = ux$, 则 $x(1-x)u'' + (x^2 - 2x + 2)u' = 0$.

令 $z = u'$, 则 $x(1-x)z' + (x^2 - 2x + 2)z = 0$.

可以解得通解 Y .

化为常系数线性方程: $2y' + py = 0$.

则 $y_1 = (e^{-\int \frac{p}{2} dx})$, $y_1' = -\frac{p}{2} y_1 + (-\frac{p}{2})^2 y_1$

故 $y_1' + p y_1 + q y_1 = -\frac{1}{4}(2p' + p^2 - 4q) e^{-\int \frac{p}{2} dx} \equiv C y_1$.

故 $2p' + p^2 - 4q = -4C$.

令 $y = y_1 u$, 则 $u'' + cu = 0$.

故 $y = u e^{-\int \frac{p}{2} dx}$

例: $(x^2 - 1)y'' + 4xy' + 2y = 0$.

解: $y'' + \frac{4x}{x^2 - 1} y' + \frac{2}{x^2 - 1} y = 0$

则 $p = \frac{4x}{x^2 - 1}$, $q = \frac{2}{x^2 - 1}$, $p' = \frac{4}{x^2 - 1} - \frac{8x^2}{(x^2 - 1)^2}$

$$\text{故 } 2p' + p^2 - 4q = \frac{8}{x^2-1} - \frac{16x^2}{(x^2-1)^2} + \frac{16x^2}{(x^2-1)^2} - \frac{8}{x^2-1} = 0.$$

故方程是特殊系数方程.

$$\text{令 } y = uv, \text{ 则 } v = e^{-\int \frac{2x}{x^2-1} dx} = e^{-\ln|x^2-1|} = \frac{1}{x^2-1}$$

而 $u'' + 0 = 0$, 故 $u = C_1 + C_2 x$.

$$\text{得 } y = \frac{C_1 + C_2 x}{x^2 - 1}.$$

2. 常数变易法

设二阶方程 $y'' + p_1(x)y' + p_2(x)y = f(x)$

已知对应齐次方程的通解 $Y = C_1 y_1(x) + C_2 y_2(x)$.

变易常数, 得 $y = u_1 y_1 + u_2 y_2$.

$$y' = u_1 y_1' + u_1' y_1 + u_2 y_2' + u_2' y_2, \text{ 令 } u_1' y_1 + u_2' y_2 = 0$$

$$\text{则 } y'' = u_1 y_1'' + u_1' y_1' + u_2 y_2'' + u_2' y_2'$$

回代方程, 得 $u_1' y_1'(x) + u_2' y_2'(x) = f(x)$

$$\text{故可得方程组 } \begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix} \begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \begin{bmatrix} 0 \\ f(x) \end{bmatrix}$$

利用Cramer法则可解出 $u_1' = -\frac{y_2(x)}{W(x)} f(x)$, $u_2' = -\frac{y_1(x)}{W(x)} f(x)$.

其中 W 是朗斯基行列式.

例: 求 $y'' - 2y' + y = 2xe^x$

解: 通解 $Y = (C_1 + C_2 x) e^x$

$$\text{解方程 } \begin{bmatrix} e^x & xe^x \\ e^x & (x+1)e^x \end{bmatrix} \begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \begin{bmatrix} 0 \\ f(x) \end{bmatrix}$$

$$u_1' = -2x^2, u_2' = 2x$$

故通解 $y = (-\frac{1}{3}x^3 + C_1 x + C_2) e^x$.

例: 求 $(x^2-1)y'' + 4xy' + 2y = 6x$.

解: 对应的齐次方程 $(x^2-1)y'' + 4xy' + 2y = 0$

$$\text{对应的通解为 } Y = \frac{C_1}{x^2-1} + \frac{C_2 x}{x^2-1} = \frac{C_1}{x-1} + \frac{C_2}{x+1}$$

设原方程的解为 $y = \frac{u_1}{x-1} + \frac{u_2}{x+1}$.

$$\text{故} \begin{bmatrix} \frac{1}{x-1} & \frac{1}{x+1} \\ -\frac{1}{(x-1)^2} & -\frac{1}{(x+1)^2} \end{bmatrix} \begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{6x}{2x^2-1} \end{bmatrix}$$

可以解出 u_1' 与 u_2' .

或可求出特解 $y^* = x$, 利用常数变易法求解.

$$\text{例: } (1-x)y'' + xy' - y = 0$$

解: 特解 $y^* = x$, 故解为 $y = C_1x + C_2e^x$.

$$\text{例: } (1-x)y'' + xy' - y = -(1-x)^2.$$

解: 设 $y = u_1x + u_2e^x$

$$\text{故} \begin{bmatrix} x & e^x \\ 1 & e^x \end{bmatrix} \begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \begin{bmatrix} 0 \\ x-1 \end{bmatrix}$$

$$\text{故 } u_1' = -1, \quad u_2' = xe^{-x}, \quad u_1 = C_1 - x, \quad u_2 = C_2 - (x+1)e^{-x}.$$

$$\text{代入, 得 } y = C_1x - x^2 + C_2e^x - (x+1) = -x^2 - x - 1 + C_1x + C_2e^x.$$

3. 幂级数法

设 $y^* = \sum a_n x^n$, 利用待定系数法可以求出特解.

§3 微分方程组与线性微分方程组

向量形式: $\frac{d\vec{x}}{dt} = \vec{f}(t, \vec{x}), \quad \vec{x}|_{t=t_0} = \vec{x}_0$.

标准形式: $\frac{d\vec{x}}{dt} = A(t) \cdot \vec{x} + \vec{f}(t)$

线性方程组一般理论:

① 解空间是线性空间

通解: $\vec{x} = [\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n] [C_1, C_2, \dots, C_n]^T$.

称 $[\vec{x}_1, \dots, \vec{x}_n]$ 为基础解系 (线性无关)

② 朗斯基行列式

$$W(t) = |\vec{x}_1 \quad \vec{x}_2 \quad \dots \quad \vec{x}_n|$$

当 $W(t) \neq 0$ 时, $\vec{x}_i$ 是线性无关的.

常系数的解法 ($A(t) = A$)

1. 齐次情况 ($\vec{f}(t) = \vec{0}$)

设 $\vec{x}(t) = \vec{v} e^{\lambda t}$, 则得 $A\vec{v} = \lambda\vec{v}$

故 λ 是 A 的特征值, $\vec{v}$ 是 λ 对应的一个特征向量.

解法: ① 解 $|A - \lambda E| = 0$, 得特征值 $\lambda_1, \dots, \lambda_n$

② 找 λ_i 对应的特征向量 $\vec{v}_i$

方程的通解即为 $y = \sum_{i=1}^n C_i \vec{v}_i e^{\lambda_i t}$.

例:
$$\begin{cases} x' = -3x + 4y - 2z \\ y' = x + z \\ z' = 6x - 6y + 5z \end{cases}$$

解:
$$A = \begin{bmatrix} -3 & 4 & -2 \\ 1 & 0 & 1 \\ 6 & -6 & 5 \end{bmatrix}$$

解特征方程 $|A - \lambda E|$, 得 $(-\lambda+2)(\lambda-1)(\lambda+1) = 0$.

故 $\lambda_1 = 2, \lambda_2 = 1, \lambda_3 = -1$.

解对应的特征向量 (可取矩阵任意两行的外积)

取 $\vec{v}_1 = (0, 1, 2)^T$, $\vec{v}_2 = (1, 1, 0)^T$, $\vec{v}_3 = (1, 0, -1)^T$.

故方程对应的通解为 $(x, y, z)^T = (0, 1, 2)^T e^{2t} + (1, 1, 0)^T e^t + (1, 0, -1)^T e^{-t}$.

例:
$$\begin{cases} x' = x \\ y' = 2x + y - 2z \\ z' = 3x + 2y + z \end{cases}$$

解:
$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & -2 \\ 3 & 2 & 1 \end{bmatrix}$$

解特征方程 $|A - \lambda E| = 0$, 得 $\lambda_1 = 1, \lambda_2 = 1+2i, \lambda_3 = 1-2i$.

解对应的特征向量 $\vec{v}_1 = (2, -3, 2)^T$, $\vec{v}_2 = (0, 1, -i)^T$

$$\vec{x} = \vec{v}_2 e^{\lambda_2 t} = (0, 1, -i)^T e^{(1+2i)t} = e^t (0, \cos 2t, \sin 2t)^T + ie^t (0, \sin 2t, -\cos 2t)^T.$$

故通解为 $C_1(2, -3, 2)^T e^{2t} + C_2 e^t (0, \cos 2t, \sin 2t)^T + C_3 e^t (0, \sin 2t, -\cos 2t)^T$.

2. 非齐次情况

① 消元法

例:
$$\begin{cases} x' = x + 2y + e^t & ① \\ y' = 4x + 3y & ② \end{cases}$$

解: 由方程①得 $y = \frac{1}{2}x' - \frac{1}{2}x - \frac{1}{2}e^t$.

代入方程②得 $\frac{1}{2}x'' - \frac{1}{2}x' - \frac{1}{2}e^t = 4x + \frac{3}{2}x' - \frac{3}{2}x - \frac{3}{2}e^t$ ③

可以求得 x 关于 t 的方程, 代入可解得 y 关于 t 的方程.

方程③可化为 $x'' - 4x' - 5x = -2e^t$.

特征方程 $\lambda^2 - 4\lambda - 5 = 0$, $\lambda_1 = 5$, $\lambda_2 = -1$.

齐次通解 $x = C_1 e^{-t} + C_2 e^{5t}$.

设特解 $x^* = A e^t$, 得 $-8A = -2$, $A = \frac{1}{4}$.

故 $x = x + x^* = C_1 e^{-t} + C_2 e^{5t} + \frac{1}{4} e^t$.

$y = -C_1 e^t + 2C_2 e^{5t} - \frac{1}{2} e^t$.

3. 特征值有重根

设 λ 是 k 重特征根

方程 $AX' = X$, $|A - \lambda E| = 0$ 且 λ 是 k 重根

猜测 $\vec{x} = (\vec{v}_0 + t\vec{v}_1 + \dots + \frac{t^{k-1}}{(k-1)!} \vec{v}_{k-1}) e^{\lambda t}$

对 t 求导, 得 $\frac{d\vec{x}}{dt} = (\vec{v}_1 + t\vec{v}_2 + \dots + \frac{t^{k-2}}{(k-2)!} \vec{v}_{k-1}) e^{\lambda t} + \lambda (\vec{v}_0 + t\vec{v}_1 + \dots + \frac{t^{k-1}}{(k-1)!} \vec{v}_{k-1}) e^{\lambda t}$
 $= (A\vec{v}_0 + tA\vec{v}_1 + \dots + \frac{t^{k-1}}{(k-1)!} A\vec{v}_{k-1})$

故 $\vec{v}_1 + \lambda\vec{v}_0 = A\vec{v}_0$, $\vec{v}_2 + \lambda\vec{v}_1 = A\vec{v}_1$, $\dots$, $0 + \lambda\vec{v}_{k-1} = A\vec{v}_{k-1}$

$\Rightarrow \vec{v}_1 = (A - \lambda E)\vec{v}_0$, $\vec{v}_2 = (A - \lambda E)\vec{v}_1$, $\dots$, $(A - \lambda E)\vec{v}_{k-1} = 0$.

$\Rightarrow \vec{v}_m = (A - \lambda E)^m \vec{v}_0$, $\vec{v}_k = (A - \lambda E)^{k-1} \vec{v}_0 = 0$.

求 $(A - \lambda E)^k \vec{v}_0 = 0$, 有 $\vec{v}_0, \dots, \vec{v}_0^k$ 这 k 个线性无关的向量.

取 $\vec{x}_1 = (\vec{v}_0 + t(A - \lambda E)\vec{v}_0 + \frac{t^2}{2!} (A - \lambda E)^2 \vec{v}_0 + \dots + \frac{t^{k-1}}{(k-1)!} (A - \lambda E)^{k-1} \vec{v}_0) e^{\lambda t}$

其他 $\vec{x}_i$ 同理

例:
$$\begin{cases} x' = -x + y \\ y' = -y + 4z \\ z' = x - 4z \end{cases}$$

解: 对应的矩阵为
$$\begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 4 \\ 1 & 0 & -4 \end{bmatrix}$$

解方程 $|A - \lambda E| = 0$, 得 $-\lambda(\lambda+3)^2 = 0$

故 A 的特征值 $\lambda_1 = 0, \lambda_2 = \lambda_3 = -3$.

对于 $\lambda = 0$, $\vec{v}_1 = (4, 4, 1)$, 故 $\vec{x}_1 = (4, 4, 1)$.

对于 $\lambda = -3$, $(A - \lambda E)^2 \vec{v} = 0$, $(A - \lambda E)^2 = \begin{bmatrix} 4 & 4 & 4 \\ 4 & 4 & 4 \\ 1 & 1 & 1 \end{bmatrix}$

取 $\vec{v}_2 = (1, 0, -1)$, $\vec{v}_3 = (0, 1, -1)$.

$\vec{x}_2 = (\vec{v}_2 + t(A - \lambda E)\vec{v}_2) e^{-3t} = ((1, 0, -1) + t(2, -4, 2)) e^{-3t} = (1+2t, -4t, -1+2t) e^{-3t}$.

$\vec{x}_3 = (\vec{v}_3 + t(A - \lambda E)\vec{v}_3) e^{-3t} = (t, 1-2t, -1+t) e^{-3t}$

故 $\vec{x} = C_1 \vec{x}_1 + C_2 \vec{x}_2 + C_3 \vec{x}_3$.

另解: $y = x' + x$,

则 $x'' + x' = -x' - x + 4z$, $z = \frac{1}{4}x'' + \frac{1}{2}x' + \frac{1}{4}x$

则 $\frac{1}{4}x''' + \frac{1}{2}x'' + \frac{1}{4}x' = x - x'' - 2x' - x$, 即 $\frac{1}{4}x''' + \frac{3}{2}x'' + \frac{9}{4}x' = 0$

化简得 $x''' + 6x'' + 9x' = 0$, 对应的解为 $x = C_1 + (C_2 + C_3 t) e^{-3t}$.

不想算矩阵可以用消元法.

4. 非齐次线性微分方程组

$$\frac{d\vec{x}}{dt} = A\vec{x} + f(t)$$

对应齐次方程的解为 $\Phi(t)$, 作变动常数, 则 $\vec{x} = \Phi(t) \vec{c}(t)$

代入, 得 $\Phi(t) \vec{c}'(t) = f(t)$, $\vec{c}'(t) = \Phi^{-1}(t) f(t)$

消元法唯一-真神.

例: $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} e^t \\ 0 \end{bmatrix}$

解: 对应齐次方程组的解 $|\lambda E - A| = 0$,

$$\lambda_1 = 5, \lambda_2 = -1.$$

对应的解为 $\vec{x}_1 = (1, 2)^T e^{5t}$, $\vec{x}_2 = (1, -1)^T e^{-t}$.

取 $\Phi(t) = \begin{bmatrix} e^{5t} & e^{-t} \\ 2e^{5t} & -e^{-t} \end{bmatrix}$

故 $\begin{bmatrix} e^{5t} & e^{-t} \\ 2e^{5t} & -e^{-t} \end{bmatrix} \begin{bmatrix} C_1'(t) \\ C_2'(t) \end{bmatrix} = \begin{bmatrix} e^t \\ 0 \end{bmatrix}$

利用 Cramer 法则得 $C_1'(t) = \frac{1}{3}e^{-4t}$, $C_2'(t) = \frac{2}{3}e^{2t}$

故 $C_1 = -\frac{1}{12}e^{-4t} + \tilde{C}_1$, $C_2 = \frac{1}{3}e^{2t} + \tilde{C}_2$

即 $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} e^{5t} & e^{-t} \\ 2e^{5t} & -e^{-t} \end{bmatrix} \begin{bmatrix} -\frac{1}{12}e^{-4t} + C_1 \\ \frac{1}{3}e^{2t} + C_2 \end{bmatrix}$

可以乘开, 但没必要.

§ 习题课

1. $\frac{dy}{dx} = \frac{y}{x(x^2y^2-1)}$, $y(1)=1$.

解: $\frac{dx}{dy} = \frac{x(x^2y^2-1)}{y} = x^3y - \frac{x}{y}$

$\Rightarrow \frac{dx}{x^3dy} + \frac{1}{x^2y} = y$, 是 Bernoulli 方程.

2. $y \cos x dx + (y+2) \sin x dy = 0$.

解: $\frac{\partial M}{\partial y} = \cos x$, $\frac{\partial N}{\partial x} = (y+2) \cos x$

$\frac{\frac{\partial N}{\partial y} - \frac{\partial M}{\partial x}}{M} = \frac{1+y}{y} = \frac{1}{\mu} \frac{d\mu}{dy}$, $\ln \mu = \ln y + y$.

取 $\mu = ye^y$, 是全微分方程.

$\Rightarrow y^2e^y \cos x dx + y(y+2)e^y \sin x dy = 0$.

3. $(1-y) \frac{dy}{dx^2} = (1 + \frac{dy}{dx}) \frac{dy}{dx}$, $y(1)=0$, $y'(1)=-1$.

解: 令 $\frac{dy}{dx} = p$, 则 $(1-y)p \frac{dp}{dy} = (1+p)p$

注意 $p=0$.

$$4. 4x^2 \frac{d^2y}{dx^2} + 8x \frac{dy}{dx} + y = 0$$

解: 是 Euler 方程, 令 $x=e^t$

$$\Rightarrow 4 \frac{d^2y}{dt^2} + 4 \frac{dy}{dt} + y = 0$$

$$\Rightarrow y = (c_1 + c_2 t) e^{-\frac{t}{2}} = (c_1 + c_2 \ln x) \cdot x^{-\frac{1}{2}}.$$

$$5. x \frac{d^2y}{dx^2} - \frac{dy}{dx} + (1-x)y = 4x^2 e^x.$$

解: 利用降阶法.

$$\text{齐次方程 } x \frac{d^2y}{dx^2} - \frac{dy}{dx} + (1-x)y = 0, \text{ 一个解为 } e^x.$$

设原方程的解为 $y = ue^x$, 代入得

$$u'' + \frac{2x-1}{x} u' = 4x, \text{ 令 } z=u', \text{ 则 } \frac{dz}{dx} + \frac{2x-1}{x} z = 4x.$$

再次用变易常数法.

$$\text{另解: } y'' - \frac{1}{x} y' + \frac{1-x}{x} y = 0, \text{ 用 Liouville 公式}$$

$$y = c_1 y_1 + c_2 y_2, \int \frac{1}{y_1^2} e^{-\int p(x) dx} dx$$

$$\text{设 } y = u_1 y_1 + u_2 y_2, \text{ 则 } \begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix} \begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \begin{bmatrix} 0 \\ 4xe^x \end{bmatrix}$$

又难点: 猜特解.

$$b. \text{ 设 } a, b, c > 0, ay'' + by' + cy = f(x) \text{ 有解 } y_1(x), y_2(x)$$

$$\text{证明: } \lim_{x \rightarrow +\infty} (y_1 - y_2) = 0.$$

$$\text{证明: 令 } \varphi = y_1 - y_2, \text{ 则 } a\varphi'' + b\varphi' + c\varphi = 0.$$

$$\text{特征方程 } a\lambda^2 + b\lambda + c = 0, \text{ 得 } \lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\textcircled{1} b^2 - 4ac > 0, \text{ 则 } \lambda_2 < \lambda_1 < 0, \varphi = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}.$$

$$\text{故 } \lim_{x \rightarrow +\infty} \varphi(x) = 0$$

$$\textcircled{2} b^2 - 4ac = 0, \text{ 则 } \lambda_1 = \lambda_2 = -\frac{b}{2a} < 0, \varphi = (c_1 + c_2 x) e^{-\frac{b}{2a} x}.$$

$$\text{故 } \lim_{x \rightarrow +\infty} \varphi(x) = 0.$$

$$\textcircled{3} b^2 - 4ac < 0, \text{ 则 } \varphi = e^{-\frac{b}{2a} x} (c_1 \cos \beta x + c_2 \sin \beta x), \beta = \sqrt{4ac - b^2}.$$

$$\text{由于 } \sin \beta x \text{ 与 } \cos \beta x \text{ 是有界量, 故 } \lim_{x \rightarrow +\infty} \varphi(x) = 0.$$

綜上, $\lim_{x \rightarrow +\infty} (y_1 - y_2) = 0$.

$$7. \begin{cases} x' = -5x - 10y - 20z \\ y' = 5x + 5y + 10z \\ z' = 2x + 4y + 9z \end{cases}$$

解: 解方程 $|A - \lambda E| = 0$, $\lambda_1 = 5$, $\lambda_2 = 2+i$, $\lambda_3 = 2-i$

$\lambda = 5$ 时, $(A - \lambda E)\vec{v}_1 = 0$, $\vec{v}_1 = (-2, 0, 1)^T$.

$\lambda = 2+i$ 时, $(A - \lambda E)\vec{v}_2 = 0$, $\vec{v}_2 = (-3-i, -2+i, 2)^T$.

$\vec{v}_2 \cdot e^{\lambda t} = e^{2t}(-3\cos t + \sin t, -2\cos t - \sin t, 2\cos t) + i e^{2t}(-3\sin t - \cos t, \cos t - 2\sin t, 2\sin t)$.

故 $\vec{x} = C_1(-2, 0, 1)^T e^{5t} + C_2 e^{2t}(-3\cos t + \sin t, -2\cos t - \sin t, 2\cos t)^T + C_3 e^{2t}(-3\sin t - \cos t, \cos t - 2\sin t, 2\sin t)^T$.

$$8. \begin{cases} x' = 4x - 3y + \sin t \\ y' = 2x - y - 2\cos t \end{cases}, \text{ 且 } x(0) = 0, y(0) = 0$$

解: 采用消元法, $y = -\frac{1}{3}x' + \frac{4}{3}x + \frac{1}{3}\sin t$

$$\Rightarrow -\frac{1}{3}x'' + \frac{4}{3}x' + \frac{1}{3}\cos t = 2x - y - 2\cos t.$$

9. 求 $f(x)$, 使得 $f(x) + f'(-x) = x^2$.

解: $f(-x) + f'(x) = x^2$

求导, 得 $-f'(-x) + f''(x) = 2x$.

相加得 $f''(x) + f(x) = x^2 + 2x$, 定解条件 $f(0) + f'(0) = 0$.